

Software Requirements Specification (SRS)

Project: GrievanceConnect - Smart Hierarchical Student Grievance Management System

1. Introduction

GrievanceConnect is a web-based application that enables students to submit grievances either anonymously or with their identity. Complaints are routed to the appropriate authority and automatically escalated if unresolved within defined timelines.

2. Objectives

- Secure grievance submission
- Anonymous reporting
- Hierarchical escalation
- Transparent status tracking
- Role-based access control

3. Users

Student, Staff, HOD, Principal, Administrator

4. Functional Requirements

- User registration/login
- JWT authentication and role-based authorization
- Complaint submission with attachments
- Anonymous complaint option
- Complaint status tracking
- Complaint history
- Automatic escalation
- Notifications
- Dashboard and reports
- Feedback after resolution

5. Non-Functional Requirements

Security using Spring Security and BCrypt, performance with REST APIs, responsive UI, auditability, maintainability, and scalability.

6. Technology Stack

Frontend: React.js

Backend: Spring Boot (Gradle)

Database: PostgreSQL

ORM: Spring Data JPA

Security: Spring Security + JWT

IDE: IntelliJ IDEA and

Antigravity

7. Database Tables

Roles, departments, users, complaints, complaint_history, escalations, attachments, notifications

8. Future Enhancements

AI-based categorization, priority detection, sentiment analysis, duplicate complaint detection, email/SMS notifications.